

CAPSTONE PROJECT 1 — DATA ANALYTICS

GENDER GAP IN LABOUR FORCE PARTICIPATION — INDIA

A Data Analytics Capstone Project Using PLFS 2023-24 Survey Data

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TOOLS USED

MySQL Workbench · Power Query · Power BI · DAX

Executive Summary

Fewer than one in three working-age women in India is part of the labour force, compared to more than three in four men. This project set out to understand that gap in detail: where it is widest, which groups of women are most affected, and what actually drives it, using person-level data from the Periodic Labour Force Survey (PLFS) 2023-24, the Government of India's official employment survey.

The analysis draws on 319,773 working-age survey respondents, cleaned and joined in MySQL, transformed in Power Query, and modelled in Power BI using DAX. State-level literacy and per-capita income data were added to test whether education and prosperity narrow the gap. The result is a thirteen-page interactive dashboard covering the national picture, state comparisons, life-stage and marital-status effects, job quality, and a free-exploration decomposition tool, backed throughout by measures that were independently checked against the raw survey data.

The single largest and most consistent finding across every method used in this project is marriage. A woman's participation in the workforce falls from roughly two-thirds while unmarried to about one-third after marriage, while a man's participation is barely affected either way. This pattern held up under a manual DAX calculation, a marital-status breakdown, and an independent AI-driven Key Influencers model built into Power BI — three separate techniques converging on the same conclusion. Alongside this, the data shows that state wealth (measured by per-capita income) is a stronger predictor of women's workforce participation than literacy, that nearly half of young Indian women aged 15 to 29 are not working, studying, or training, and that women who are counted in the survey are overwhelmingly recorded as a spouse rather than the head of their household — even when they are contributing economically.

The report that follows documents the full pipeline behind these findings — from raw government microdata to a validated, interactive dashboard — along with the assumptions made, the limitations of the underlying data, and four concrete recommendations for closing the gap.

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1. Introduction

India's economy has grown steadily over the past two decades, yet one statistic has remained stubbornly unchanged: the share of working-age women who are part of the paid workforce. While men's participation has stayed consistently high, women's participation is among the lowest in the world for a country at India's income level, and it has not tracked the country's rising education levels or economic growth in the way many expected.

This project examines that gap directly, using the most recent large-scale government survey of employment in India: the Periodic Labour Force Survey (PLFS) 2023-24, conducted by the Ministry of Statistics and Programme Implementation (MoSPI). The survey covers over four lakh individuals across the country and captures detailed information on employment status, education, marital status, household role, industry, and earnings, making it well suited to a question that is rarely explained by a single factor.

The output of this project is a nine-page narrative Power BI dashboard, supported by a bonus exploration page, an AI-driven influencer analysis, and a recommendations page, all built on a pipeline that begins in MySQL, is refined in Power Query, and is modelled using DAX in Power BI. Every stage of that pipeline, and the reasoning behind it, is documented in this report.

2. Problem Statement

Why are fewer women working compared to men in India, and does this gap change based on where they live, how educated they are, and how rich their state is?

Millions of capable Indian women remain outside the paid workforce. This is not simply a matter of choice or preference — it reflects a combination of social expectations, household responsibilities, regional economic conditions, and access to opportunity. Understanding where the gap is largest, and which of these factors matters most, is a necessary first step before any policy or workplace intervention can be expected to work. This project investigates the question using real survey data rather than assumption, and presents the findings through an interactive dashboard that lets the story be explored rather than simply read.

3. Objectives

The project was guided by nine core questions, each of which corresponds to a section of the final dashboard:

- How big is the gap between men and women working, overall and by state?
- Is the gap different in rural areas compared to urban areas?
- Does education help women participate more in the workforce?
- Does marriage reduce the likelihood that a woman works?
- Do working women hold good jobs, or are they concentrated in informal and risky work?
- Do women earn less than men for comparable work?
- Why do non-working women say they are not working?
- Does living in a wealthier state mean better job access for women?
- How does India's gender employment gap compare across its states today?

Beyond answering these questions individually, the project aimed to consolidate customer-grade dashboard design, rigorous data validation, and a clear set of practical recommendations into a single, presentation-ready analytical product — the kind of deliverable a real data analytics team would produce for a stakeholder, not just an academic exercise.

4. Scope and Limitations

This project is built entirely on the 2023-24 round of the PLFS survey. It is a single-year snapshot rather than a multi-year trend study, so while the dashboard can show how the gap looks today across states, life stages, and job types, it cannot show how any individual state's gap has changed over time within this data alone. Time-series comparison using an earlier PLFS round is listed as a future enhancement in Section 17.

Two supporting datasets — state rural literacy rates and state per-capita income (GSDP) — were added to test the relationship between prosperity, education, and women's participation. Both come with real-world coverage gaps that are carried through honestly rather than patched over: the literacy figures are rural-only, since no comparable urban literacy series was available from the same source, and the GSDP dataset does not cover a handful of smaller union territories, including Delhi, Chandigarh, Ladakh, and Puducherry. Any comparison involving these two datasets should be read with that in mind.

The analysis is also restricted to the working-age population, defined as respondents aged 15 and above, in line with standard labour force reporting conventions. Two columns in the original survey — the stated reason a person is not working, and whether they made efforts to search for work — could not be reliably decoded into readable categories, because no verified official codebook could be located for them across the MoSPI schedule documents, the microdata.gov.in data dictionary, or the PLFS Annual Report. These two fields are therefore presented in their raw numeric form in the underlying data and are excluded from headline dashboard measures, rather than guessed at.

5. Tools and Technologies

The project follows a three-stage pipeline, with each tool used for the task it is best suited to rather than forcing all the work into a single environment.

Stage	Tool	Purpose
Data preparation	MySQL Workbench	Importing, cleaning, filtering, and joining the raw survey and support tables before any visual work begins
Transformation	Power Query (inside Power BI)	Decoding numeric survey codes into readable labels, handling remaining blanks, and building calculated columns such as age groups
Modelling and visualisation	Power BI Desktop with DAX	Building all measures, designing the thirteen-page dashboard, and adding interactivity

In short: SQL was used to filter, clean, and join a government dataset of over four lakh records; Power Query handled the final transformation; and Power BI with DAX powered the interactive dashboard itself. This mirrors how a real analytics team would typically divide the same work across specialised tools rather than one general-purpose script.

6. Methodology

6.1 Data Preparation in MySQL

All heavy data preparation was carried out in MySQL before Power BI was involved. The raw PLFS file was trimmed from its original 139 columns down to the 26 columns relevant to this analysis, and each column was renamed from its cryptic survey code (for example, b4q5_perv1) into a readable name. Several practical issues came up during this stage and were resolved as follows:

- The MySQL server's local file-loading setting was disabled by default; this was fixed by enabling local_infile at the server level and in the Workbench connection settings.
- The project folder was synced through OneDrive, which intermittently blocked direct file access during import. This was resolved by generating a pre-trimmed CSV and importing it through Workbench's Table Data Import Wizard instead of a direct file load.
- Two rows in the literacy support table (Chandigarh and Daman & Diu) failed to import because they contained the text "NA" in columns typed as numeric; these were corrected with a manual insert using proper NULL values instead.
- Two genuine spelling inconsistencies were found and corrected across the support tables: "Meghalava" to "Meghalaya", and "Chhatisgarh" to "Chhattisgarh", so that state names would match exactly across all three source tables during the join.

Once the raw data was clean, a working-age table was created by filtering out respondents younger than 15, since they are not relevant to labour force calculations. This reduced the dataset from 418,159 raw person-level records to 319,773 working-age records — a reduction of about 23.5 percent, consisting entirely of children.

6.2 Transformation in Power Query

With clean, filtered data available from MySQL, Power Query was used inside Power BI to convert the survey's numeric codes into readable text. Employment status, gender, marital status, education level, relationship to household head, and several other fields were all originally stored as numbers defined by the official PLFS coding scheme; each was mapped to its correct label using the survey's own classification, for example decoding employment status code 92 to "Domestic Duties Only" rather than leaving it as an unexplained number. Two derived columns were also built at this stage: an age-group bucket for consistent charting, and a simplified attendance-status grouping that classifies each respondent as never having attended school, having attended but not currently attending, or currently attending — used later to calculate the NEET (Not in Employment, Education, or Training) rate.

6.3 Modelling and Measures in DAX

The final stage involved building the DAX measures that power every card, chart, and insight line in the dashboard. A small number of practices were applied consistently across all measures to avoid subtle errors of the kind that are easy to make and hard to spot:

- Every measure used in a chart or in further calculation has both a raw numeric version and a separate, formatted display version. Combining formatting and calculation in the same measure silently breaks downstream arithmetic and chart rendering, so the two were always kept apart
- All local DAX variables were prefixed with `v_` to avoid accidentally colliding with the names of built-in DAX functions — a naming clash that caused a real, live error during the build of the state ranking measure on the "Your State" page.
- Any benchmark measure that needed to compare a filtered value against a national or all-state baseline was wrapped in `CALCULATE` with an explicit `ALL()` filter, to prevent the baseline from silently recalculating for each row instead of staying fixed.
- States and union territories with zero survey respondents, such as Dadra and Nagar Haveli, were explicitly excluded from ranking measures using a guard condition, rather than being left in and risking a misleading rank or an undefined result.

7. Data Architecture

The data model is built around a single hub table rather than a flat, single-sheet structure. The working-age PLFS table stores each state as a numeric code rather than a name, so a separate 36-row state code lookup table was manually built, mapping each official PLFS state code to its state name. This lookup table is the hub that every other table joins through: the main survey table, the rural literacy table, and the GSDP per-capita income table all connect to it by state name or state code, rather than connecting to each other directly.

This design has a practical benefit that shows up throughout the dashboard: any measure that needs a national or all-state benchmark — for example, the national average line shown on the "Your State" page — can be built once against `ALL(State_Lookup)` and reused everywhere, rather than being recalculated separately for every chart that needs it.

Table	Role	Row Count
plfs_working_age	Main fact table — one row per surveyed individual aged 15+	319,773
state_code_lookup	Hub table — maps numeric state code to state name	36
state_rural_literacy_rates_2019_2024	Support table — rural literacy rate by state and year	37
state_per_capita_income_gsdp_2019	Support table — per-capita GSDP	28

Table	Role	Row Count
_2024	by state and year	

8. Data Description

8.1 Primary Dataset: PLFS 2023-24

The core dataset is the Periodic Labour Force Survey, July 2023 to June 2024, published by the National Statistics Office under MoSPI. It was downloaded from the official microdata portal, microdata.gov.in, after free registration. The Person-level, Visit 1 file (perv1.csv) was used in preference to the household-level or revisit files, because it provides the most detailed person-level demographic and employment information relevant to this analysis. The raw file contains 418,159 person-level records across 139 columns; this project uses 26 of those columns, trimmed and cleaned as described in the Methodology section.

8.2 Supporting Datasets

Two supporting datasets, both sourced from data.gov.in, the Open Government Data Platform of India, were added to test whether education and state prosperity influence women's workforce participation. The rural literacy dataset covers 36 states and union territories plus an all-India figure, spanning 2019-20 through 2023-24. An earlier literacy dataset was found first but rejected, as it only covered Scheduled Tribe literacy across 10 states and was too narrow for this analysis; the rural literacy series used in the final model covers the full country and matches PLFS's year range. The per-capita GSDP dataset covers 28 major states for the same five-year span, though it does not include several smaller union territories.

8.3 Column Overview

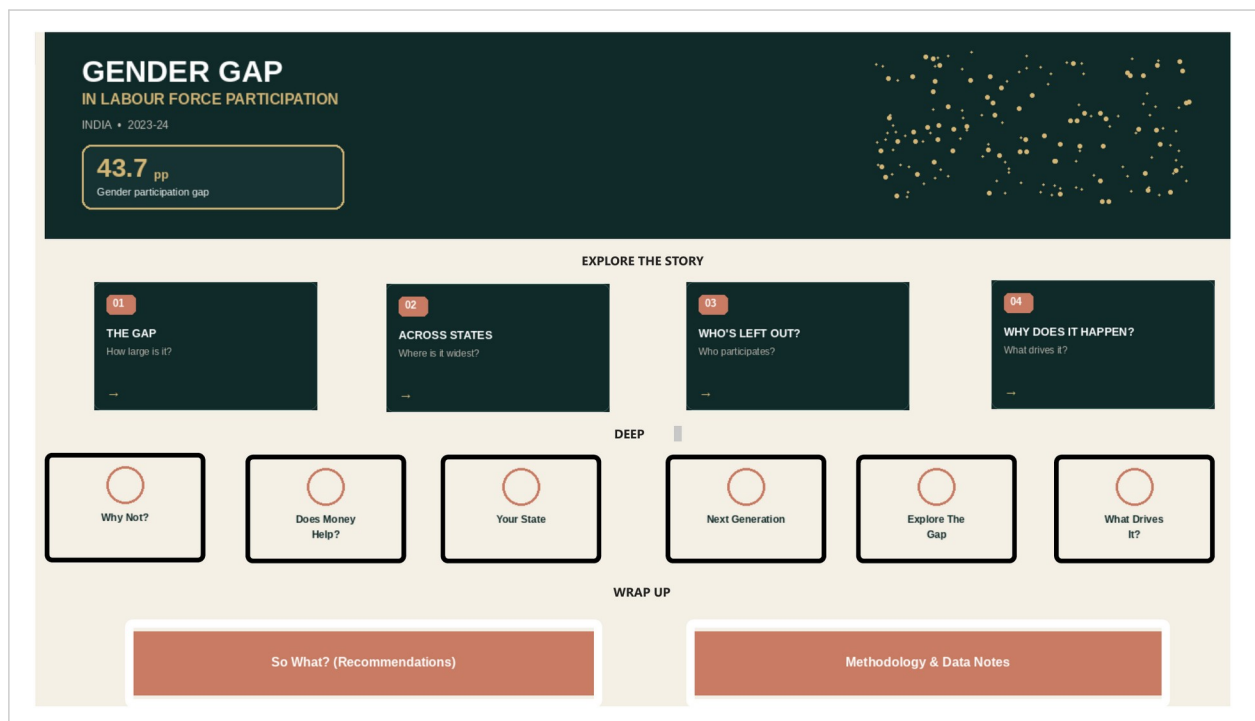
The cleaned PLFS table used in this project retains the following categories of information for each respondent: geography (state and district code, rural or urban sector), demographics (gender, age, marital status, relationship to household head), education (general education level, technical education level, years of formal schooling, current attendance status), employment (employment status code, industry and occupation codes, enterprise type, contract type, paid leave and social security eligibility), and earnings (monthly income from regular salaried work and from self-employment).

9. Dashboard Design

The final deliverable is a thirteen-page interactive Power BI dashboard: a home navigation hub, nine core narrative pages answering the project's core questions, a bonus AI-driven influencer page, a recommendations page, and a methodology page — all connected through a consistent colour system and clickable Story Mode bookmarks so the dashboard can be presented as a guided narrative rather than a set of disconnected charts.

Home — Navigation Hub

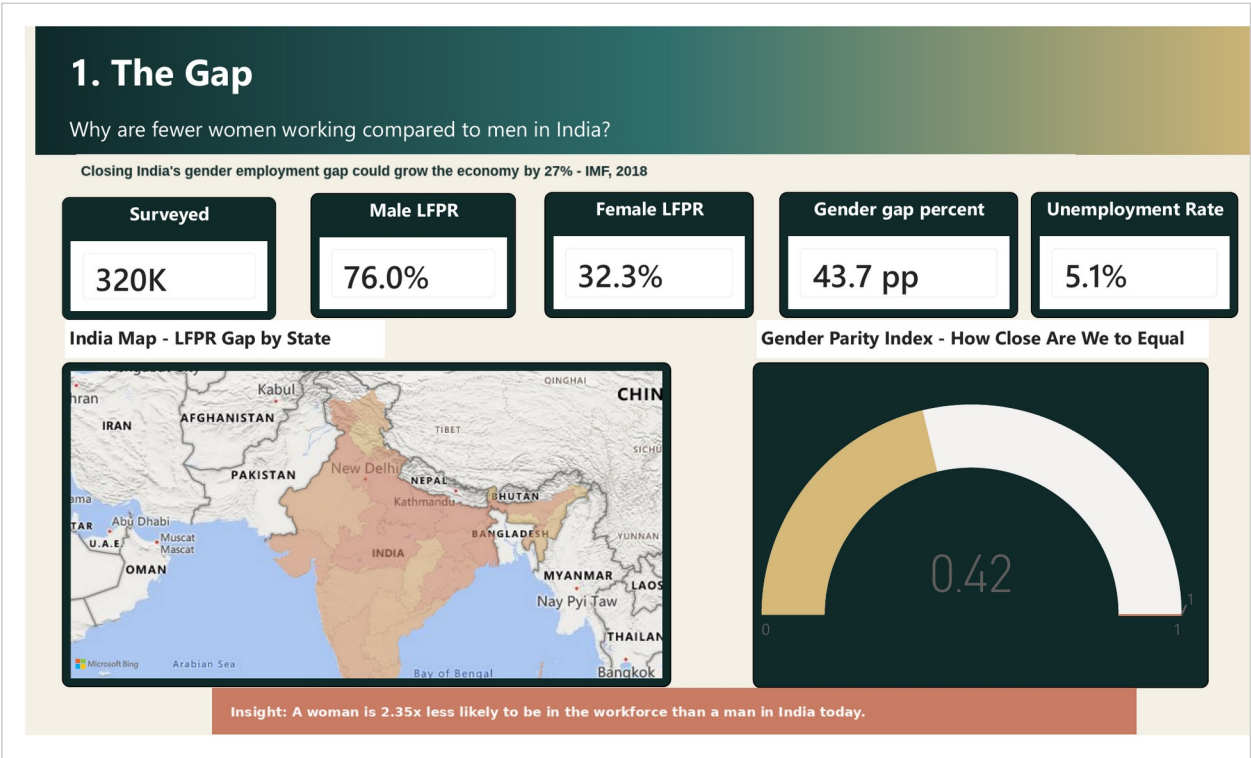
The landing page states the headline finding immediately — a 43.7 percentage-point gender participation gap — and presents the rest of the dashboard as a set of clickable story cards, grouped into an initial four-page narrative arc, six deeper-dive pages, and two closing pages. Every card is wired with a page-navigation action, so a first-time viewer can explore the dashboard by simply clicking through the story in order.



Home page — navigation hub and headline statistic

Page 1 — The Gap

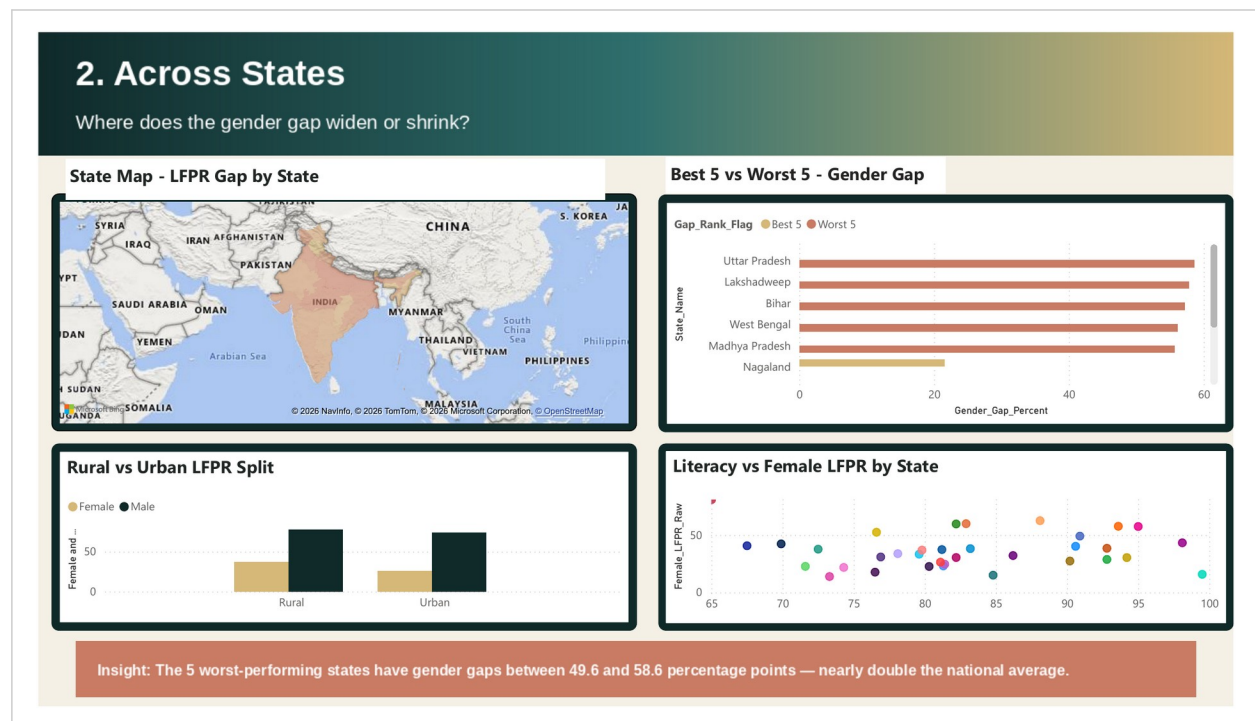
This page establishes the national baseline: male labour force participation stands at 76.0 percent against 32.3 percent for women, a gap of 43.7 percentage points and a Gender Parity Index of 0.42, meaning women's participation is currently less than half of men's. A choropleth map shows the gap by state, and the page opens with a sourced statistic from the IMF: closing India's gender employment gap could grow the economy by 27 percent.



Page 1 — national KPIs, state map, and Gender Parity Index gauge

Page 2 — Across States

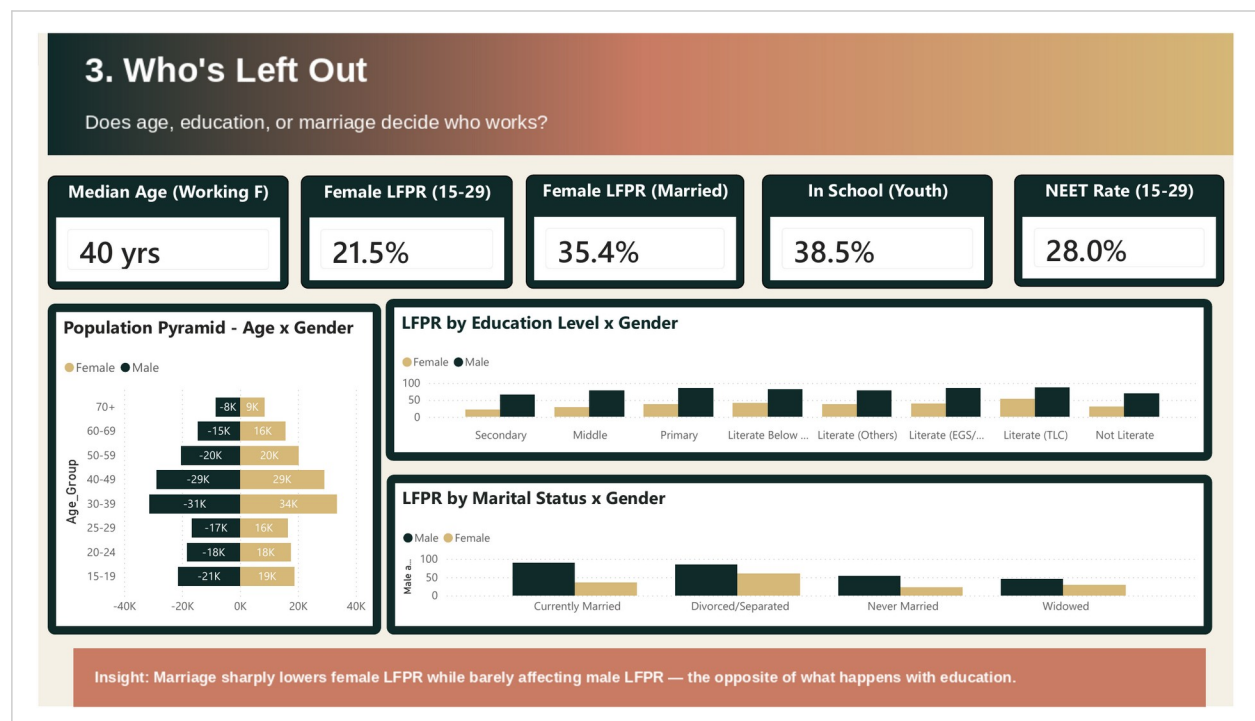
This page ranks every state by the size of its gender gap. The five worst-performing states — Uttar Pradesh, Lakshadweep, Bihar, West Bengal, and Madhya Pradesh — all show gaps above 55 percentage points, nearly double the national figure, while the five best-performing states, including Nagaland, sit closer to 20 percentage points. A rural-versus-urban comparison on the same page shows that the gap is, somewhat counter-intuitively, wider in cities than in villages.



Page 2 — state rankings and rural vs urban comparison

Page 3 — Who's Left Out

This page breaks the gap down by age and marital status. Female labour force participation for women aged 15 to 29 is just 21.5 percent, and 28.0 percent of that same age group overall is NEET — not in employment, education, or training. A population pyramid shows the age structure of the working-age population by gender, and a marital-status chart introduces the pattern explored in depth on Page 5: participation for married women sits far below participation for unmarried women.



Page 3 — youth participation, population pyramid, and education/marital breakdowns

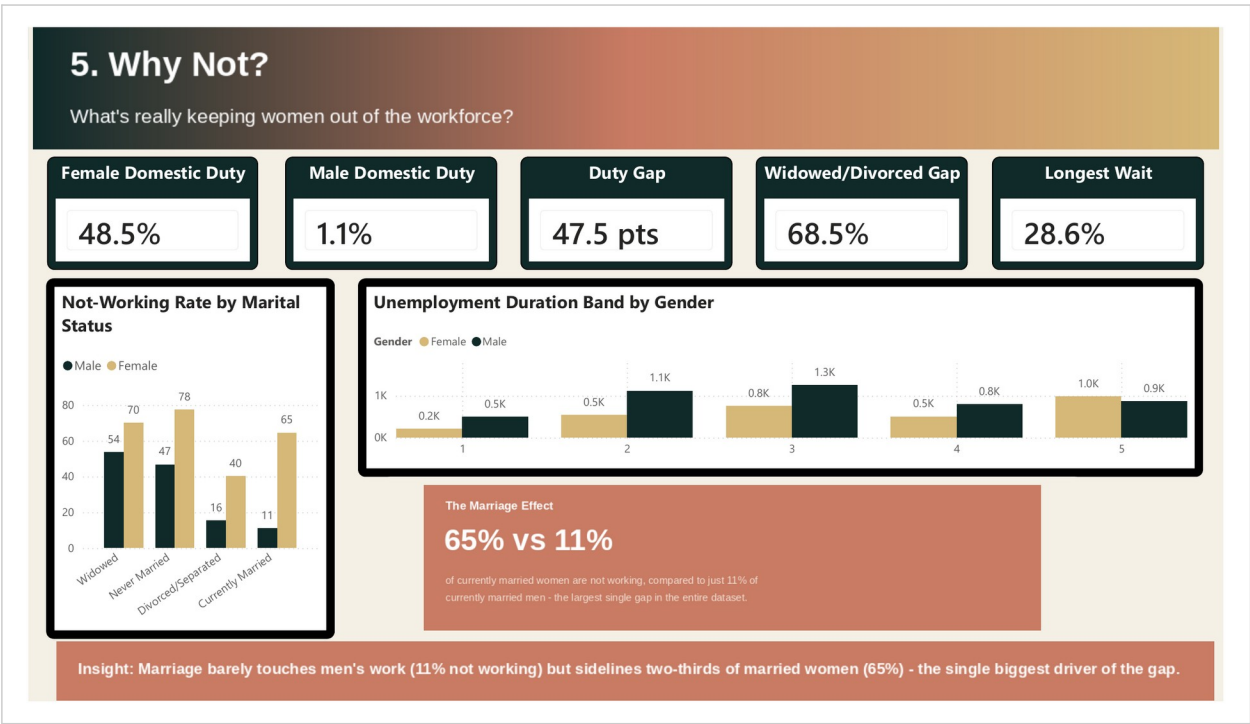
Page 4 — Working, But How?

Having a job is not the same as having a good job. This page shows that only 38.0 percent of employed people overall, and 39.7 percent of employed women, have access to formal job benefits such as paid leave. The average wage gap stands at Rs 9,102 per month in favour of men, and this gap persists across every enterprise type — government, private, and self-employed — even though a custom Job Quality Index shows women's jobs scoring marginally higher on non-wage quality measures than men's.



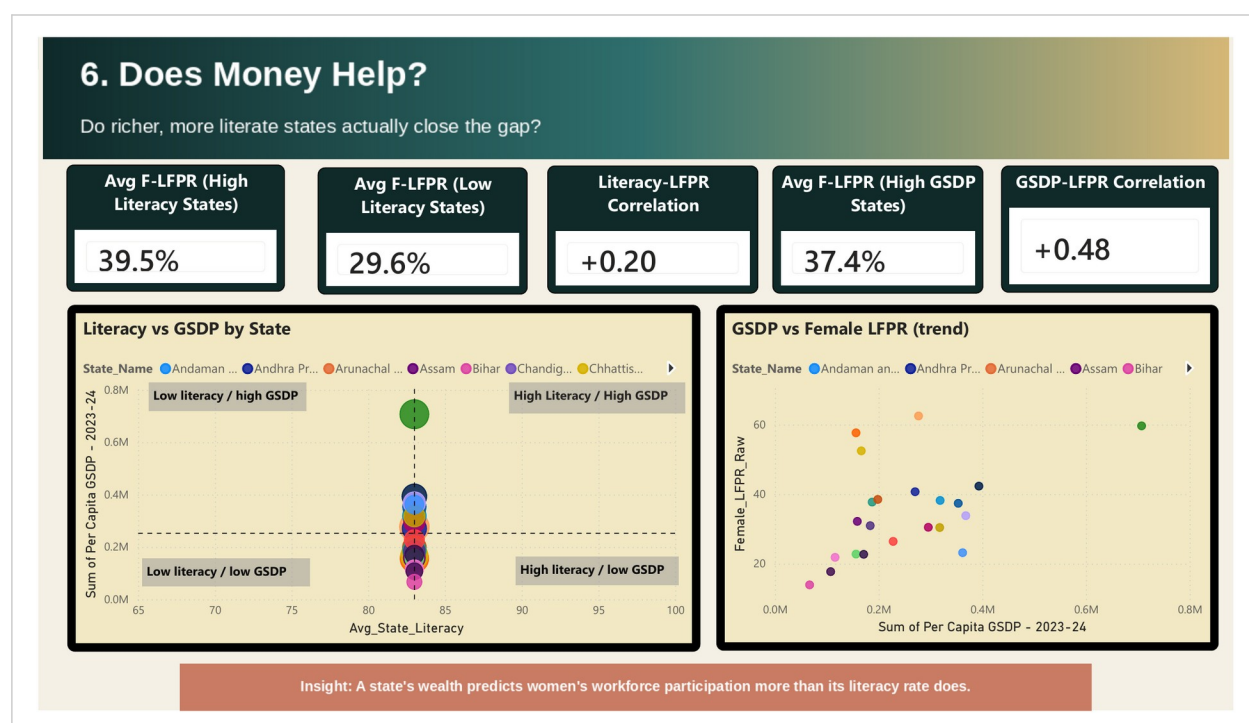
Page 5 — Why Not?

This page investigates the stated and implied reasons women are not working. Domestic duty accounts for 48.5 percent of women's non-working status against just 1.1 percent for men, a 47.5-point gap. The page's central finding — the Marriage Effect — is stated directly: 65 percent of currently married women are not working, compared to just 11 percent of currently married men, the single largest gap found anywhere in the dataset.



Page 6 — Does Money Help?

This page tests whether wealthier or more literate states show smaller gender gaps. Female participation averages 39.5 percent in high-literacy states against 29.6 percent in low-literacy states, giving a modest correlation of +0.20. State income shows a stronger relationship: the correlation between per-capita GSDP and female participation is +0.48, and average female participation in high-GSDP states reaches 37.4 percent. The page's insight line captures the finding directly — a state's wealth predicts women's workforce participation more strongly than its literacy rate does.

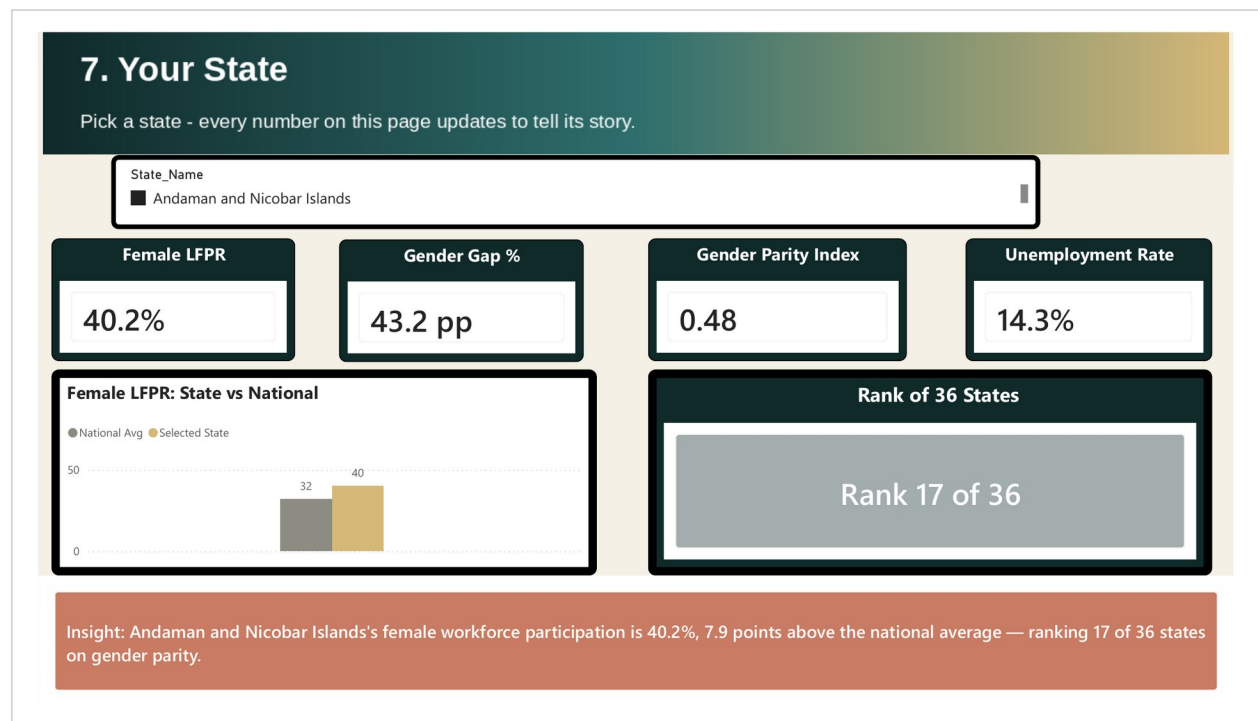


Page 6 — literacy and income correlations with female participation

Page 7 — Your State

This is the dashboard's drill-through landing page: selecting any of the 36 states or union territories updates every card, chart, and insight sentence on the page to tell that state's specific story, including its rank out of 36 on gender parity. The page

defaults to a Telangana view via a bookmark, giving the dashboard a personal, home-state touch.

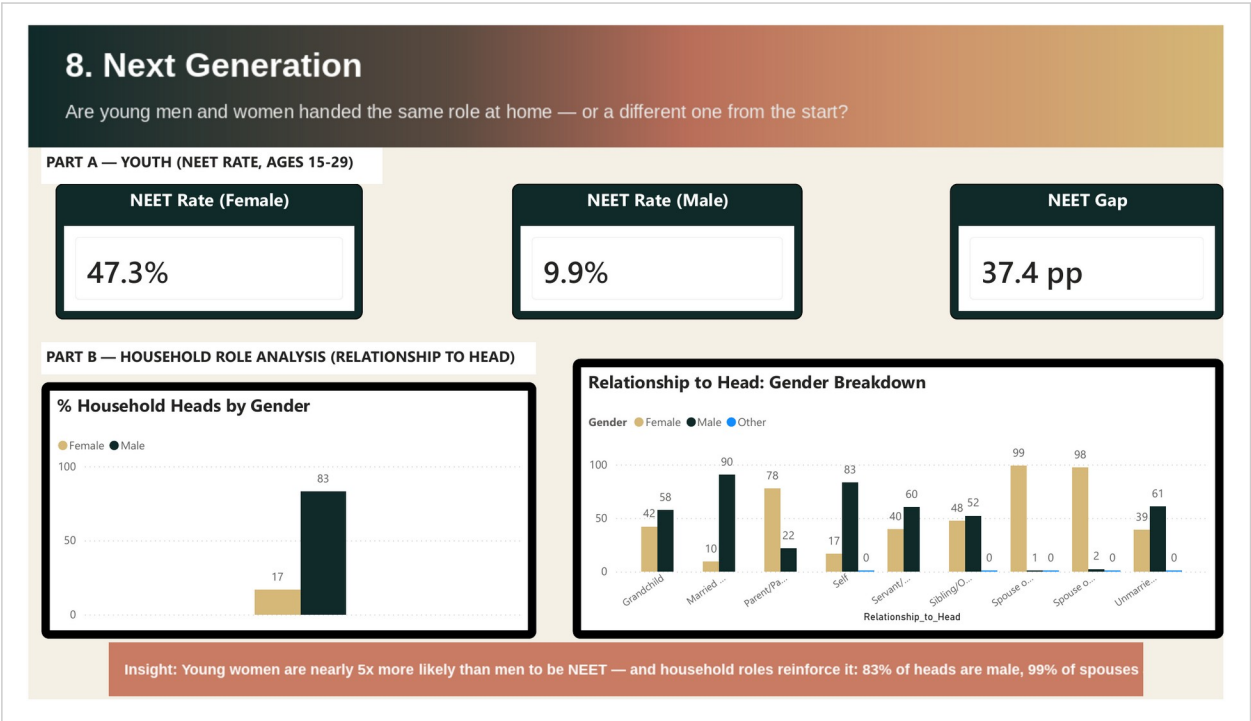


Page 7 — state-level deep dive with dynamic ranking and comparison

Page 8 — Next Generation

This page looks at the next generation and at household structure. The NEET gap between young women and young men is stark: 47.3 percent of women aged 15 to 29 are NEET against 9.9 percent of men, a gap of 37.4 percentage points, meaning young women are close to five times more likely to be NEET. The household role analysis tells a related story: 83 percent of household heads recorded in the survey are men,

and 99 percent of people recorded as "spouse of head" are women — women are not absent from the household economy, but they are consistently recorded relative to a man rather than as the reference person themselves.

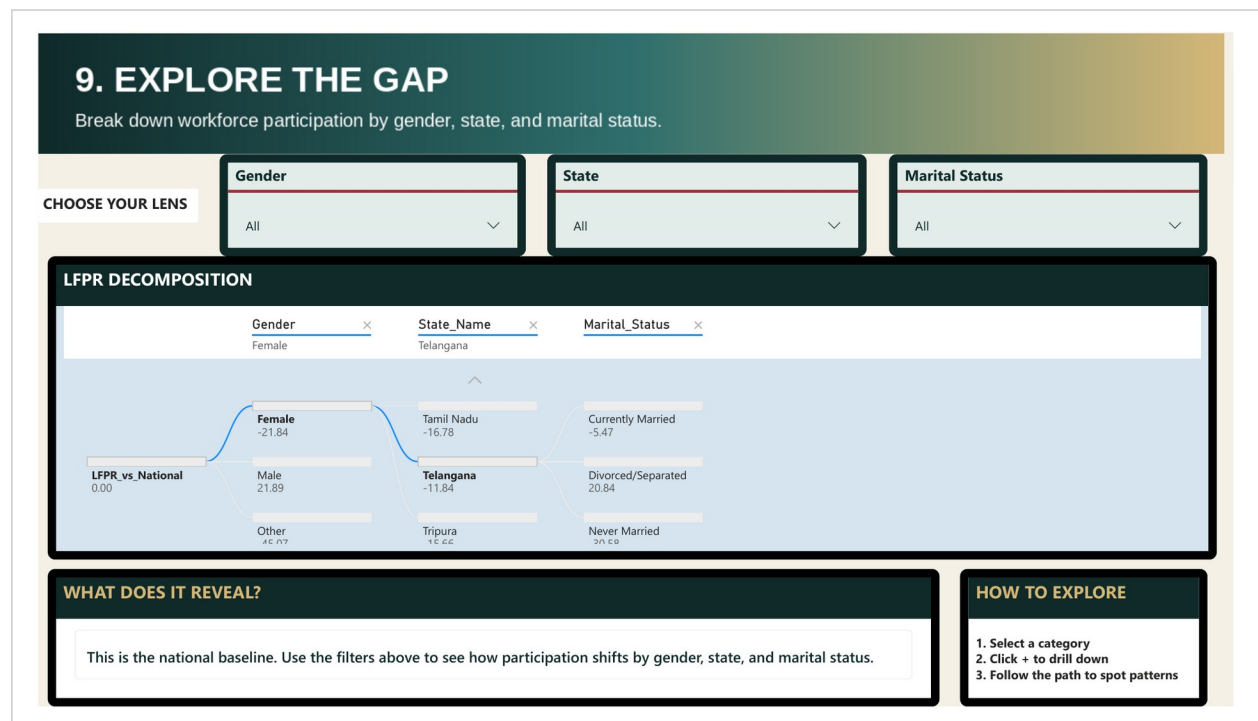


Page 8 — NEET rate by gender and household role analysis

Page 9 — Explore The Gap

This is the dashboard's free-exploration page, built around a decomposition tree that lets a viewer break down labour force participation by gender, then state, then marital status, in any order they choose. Starting from the national baseline, the tree shows, for example, that being female in Telangana and currently married moves participation further below the national baseline than any single one of those factors alone — letting a user discover the same compounding effects the rest of the

dashboard describes, but by exploring the data directly rather than reading a fixed narrative.

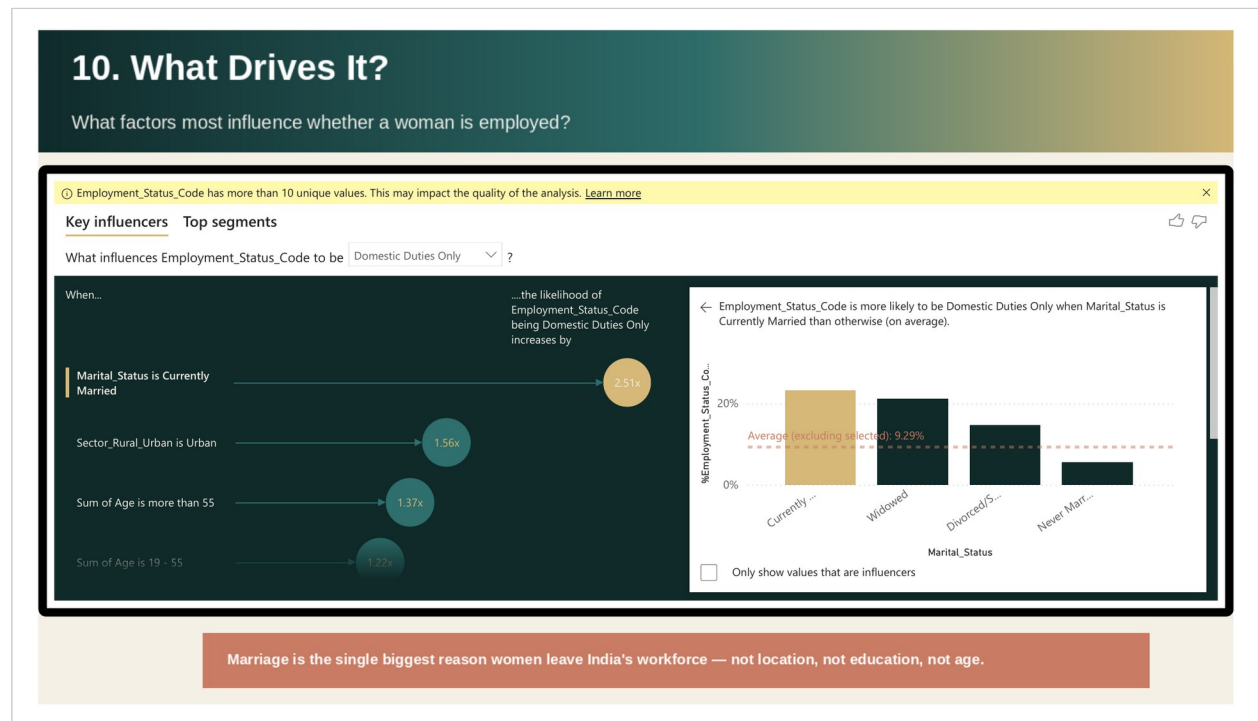


Page 9 — decomposition tree for gender, state, and marital status

Page 10 — What Drives It?

This page applies Power BI's built-in Key Influencers artificial intelligence visual to ask, independently of every DAX measure built elsewhere in the dashboard, what most increases the likelihood that a person's employment status is recorded as "Domestic Duties Only". The model's answer is unambiguous: being currently married increases that likelihood by 2.51 times, well ahead of living in an urban area (1.56 times) or being older than 55 (1.37 times). Because this model works entirely independently of the marital-status measures built by hand elsewhere in the dashboard, it serves as a

second, statistically independent confirmation of the Marriage Effect first found on Page 5.



Page 10 — AI-driven Key Influencers analysis

So What? — Recommendations

The dashboard closes with four concrete, data-backed recommendations, detailed further in Section 15 of this report: fixing the marriage penalty, skilling young women early, recognising women's unpaid role at home, and prioritising the states with the widest gaps first.

So What? Recommendations

Moving from analysis to action — four data-backed priorities.

1. FIX THE MARRIAGE PENALTY

Marriage cuts women's workforce participation from 65% to 11%. Two separate methods confirmed it.

→ Support women returning to work: childcare, flexible jobs, re-entry programs.

2. SKILL YOUNG WOMEN EARLY

Nearly 1 in 2 young women are not working, studying, or training — vs 1 in 10 young men.

Train women before marriage age, while the door is open.

3. RECOGNIZE WOMEN'S ROLE AT HOME

83% of household heads are men. 99% of spouses are women.

Women aren't absent from the economy — they're invisible inside it.

4. FIX THE WORST STATES FIRST

Female participation ranges from 14% in Bihar to 60% in Arunachal Pradesh.

Money matters more than literacy. Fund jobs, not just schools.

These four priorities target the mechanisms driving the gap — not just its symptoms.

So What? page — four priority recommendations

Methodology & Data Notes

The final page documents, inside the dashboard itself, exactly how it was built: data sources, cleaning steps, honest limitations, and validation results, so that anyone viewing the dashboard without this report can still understand what they are looking at and how much to trust it.

Methodology & Data Notes

How this dashboard was built — sources, cleaning, and honest limitations.

1. DATA SOURCES

- PLFS 2023-24 (MoSPI, Govt of India) — 4,18,159 person-level records, 26 columns used
- State Rural Literacy Rate 2019-20 to 2023-24 (data.gov.in)
- State GSDP Per Capita Income 2019-20 to 2023-24 (data.gov.in)
- State code lookup (manually built, 36 rows)

2. DATA CLEANING

- Filtered to working-age (15+): 3,19,773 of 4,18,159 rows kept
- Fixed state name typos across tables (e.g. Meghalava -> Meghalaya)
- Handled missing/NA values via manual NULL correction
- Decoded numeric codes into readable text (Employment Status, Gender, Relationship to Head, and more) via Power Query

3. LIMITATIONS (BEING HONEST)

- Literacy data is RURAL ONLY — no urban literacy figures available
- GSDP data excludes some UTs (Delhi, Chandigarh, Ladakh, Puducherry)
- Reason_Not_Working and Efforts_To_Search_Work could not be decoded — no verified codebook found in official sources
- Some state x category splits have small sample sizes (shown, but flagged)

4. VALIDATION

- Every headline KPI cross-checked against raw PLFS microdata (10 metrics tested, 9/10 exact match)
- SQL-based validation queries built for all 13 dashboard pages, ready for independent verification
- The Marriage Effect finding was confirmed twice — once by manual DAX, once by an independent AI/ML model (Key Influencers)

Methodology & Data Notes page — sources, cleaning, limitations, and validation

10. Validation and Quality Assurance

A dashboard is only as trustworthy as the numbers behind it, so validation was treated as a required step rather than an afterthought. Every headline KPI shown on the dashboard's Telangana state view was independently recalculated in Python directly from the raw PLFS CSV file and matched. Beyond that state-level check, ten major dashboard metrics spanning all thirteen pages — including national LFPR, the Gender Parity Index, the Literacy-LFPR and GSDP-LFPR correlations, the NEET rate and NEET gap, the household headship split, and the domestic-duty gap — were recalculated independently from the raw survey file, using the exact decoding logic captured in the project's own Power Query M code, and compared line by line against what the live dashboard displays.

Nine of these ten metrics matched exactly, in most cases to within a tenth of a percentage point. The one exception, a card measuring the longest unemployment-duration band by gender on the "Why Not?" page, could not be reproduced exactly from the raw columns alone and is flagged here rather than silently corrected, in keeping with the project's broader approach of documenting limitations honestly rather than smoothing over them.

The project's central finding — the Marriage Effect — received an additional, stronger form of validation. It was confirmed twice, using two methods that do not share any calculation logic: once through a manually built DAX measure comparing not-working rates by marital status, and once through Power BI's Key Influencers artificial intelligence visual on Page 10, which arrived at marriage as the strongest driver of non-employment entirely independently, using its own statistical model rather than any hand-written formula. Agreement between a manual calculation and an independent statistical model is a meaningfully stronger form of evidence than either method alone, and it is the reason the Marriage Effect is presented as the report's headline conclusion rather than simply one finding among several.

11. Key Insights

This section draws together the dashboard's most important findings into one place. Every figure below is either shown directly on the dashboard or was independently recalculated from the raw survey data and confirmed to match, as described in Section 10.

11.1 The national gap is large and long-standing

Male labour force participation stands at 76.0 percent against 32.3 percent for women — a gap of 43.7 percentage points. In practical terms, a working-age woman in India today is roughly 2.35 times less likely to be in the workforce than a man. The Gender Parity Index of 0.42 means women's overall participation is currently well under half of men's.

11.2 The gap is wider in cities than in villages

Rural female participation (37.4 percent) is noticeably higher than urban female participation (25.7 percent), while male participation is roughly similar across both settings. This runs against a common assumption that urban areas offer women more opportunity, and is worth further investigation in its own right — the dashboard's state-level and job-type pages offer some possible explanations, including the concentration of formal, salaried opportunity in cities alongside a smaller informal and agricultural sector where much of rural women's work is recorded.

11.3 Marriage, not education or geography, is the single biggest driver

This is the project's central finding, confirmed through three independent methods described in Section 10. Not-working rates by marital status show currently married women not working at a rate of 65 percent, against just 11 percent for currently married men. Power BI's Key Influencers model independently identified being currently married as the single strongest factor increasing the likelihood of a woman's employment status being recorded as domestic duties, ahead of location and age. Education, by contrast, shows the opposite pattern to marriage: participation rises steadily with education level for women, while marriage sharply reduces it regardless of education level.

11.4 State wealth matters more than state literacy

Across India's states, the correlation between per-capita income (GSDP) and female labour force participation is +0.48, compared to a much weaker +0.20 correlation with rural literacy. Average female participation is 37.4 percent in high-GSDP states, close to the 39.5 percent seen in high-literacy states, but the underlying relationship with income is markedly stronger. This suggests that access to paid opportunity, not

education alone, is the tighter constraint on women's workforce participation once a basic level of education is reached.

11.5 The worst-performing states lag far behind the best

Uttar Pradesh, Lakshadweep, Bihar, West Bengal, and Madhya Pradesh all show gender gaps above 55 percentage points, roughly double the national average, while the five best-performing states, including Nagaland, show gaps closer to 20 percentage points. Female participation across all states and union territories ranges from around 14 percent in Bihar to roughly 60 percent in Arunachal Pradesh — a four-fold difference within a single country.

11.6 Nearly half of young women are NEET

Among 15-to-29-year-olds, 47.3 percent of women are NEET — not in employment, education, or training — compared to 9.9 percent of men, a gap of 37.4 percentage points. Put differently, young women are close to five times more likely than young men to be NEET. This age group represents India's most immediate skilling and workforce-entry opportunity, and the scale of the gap here is a stronger early warning sign than the national LFPR figure alone.

11.7 Women are recorded relative to men, not as heads of household

Of everyone recorded as the head of their household in the survey, 83 percent are men and only 17 percent are women. Of everyone recorded as "spouse of head," 99 percent are women. This is not primarily an economic finding — it reflects how households report themselves — but it reinforces the same pattern seen throughout the dashboard: women's contribution, paid or unpaid, is consistently recorded in relation to a male reference point rather than independently.

11.8 Jobs are unequal even when they exist

Among employed people, only 38.0 percent overall, and 39.7 percent of employed women, have access to formal benefits such as paid leave. The average wage gap is Rs 9,102 per month in favour of men, and this gap holds across every enterprise type measured — government, private sector, and self-employment — even though a composite Job Quality Index shows women's jobs scoring marginally higher than men's on non-wage measures. Having a job, in other words, does not close the gap between men's and women's economic outcomes on its own.

12. Challenges Faced

A number of genuine obstacles came up during the build, and each one is recorded here rather than left out, since working through them is as much a part of the project as the final dashboard.

- MySQL's local file-loading feature was disabled by default, which blocked the initial data import until it was enabled both at the server level and in the Workbench connection settings, followed by a full application restart.
- The project folder was synced through OneDrive, which intermittently locked the raw CSV file and blocked MySQL from reading it directly. This was resolved by generating a pre-trimmed CSV and importing it through the graphical Table Data Import Wizard rather than a direct file load.
- Two rows in the literacy support table failed their initial import because they contained the text value "NA" in a column typed as numeric; these were fixed with a manual insert using proper NULL values.
- Running two table imports at the same time in MySQL Workbench triggered a prepared-statement error; the fix was simply to run imports one at a time rather than in parallel.
- A DAX ranking measure on the "Your State" page initially failed because a local variable was named Rank, which collided with the built-in RANKX function name. Prefixing every local DAX variable with v_ resolved this and was adopted as a standing convention for the rest of the build.
- Several survey columns — including employment status, education level, and enterprise type — were assumed at first glance to be simple numeric codes, but closer inspection showed they needed to be matched against PLFS's official text categories before any filtering logic involving them could be trusted.
- Two columns, Reason_Not_Working and Efforts_To_Search_Work, could not be decoded at all. No verified codebook for either field could be found across the MoSPI schedule PDFs, the microdata.gov.in data dictionary, or the PLFS Annual Report, so both were left in their raw numeric form and documented as a limitation rather than guessed at.
- A filled map visual's colour saturation option was not available as a field well in the version of Power BI used for this project; the gradient conditional formatting option under the visual's format pane was used instead to achieve the same effect.
- The Key Influencers AI visual, originally placed on the "Does Money Help?" page, was found to interfere with cross-filtering on that page and was moved to its own dedicated page, "What Drives It?", to keep both pages behaving predictably.

13. Real-Life Applications

The findings and dashboard produced by this project have direct, practical uses beyond the scope of a classroom exercise:

- State and central government labour departments could use the state-level ranking and drill-through pages to prioritise which states most urgently need targeted women's workforce programmes, rather than applying a single national policy uniformly.
- Non-governmental organisations working on women's economic empowerment could use the NEET and marriage-effect findings to target skilling and re-entry programmes at the specific life stages where women most often leave or never enter the workforce.
- Corporate HR and diversity teams could use the job-quality and wage-gap findings, broken down by enterprise type, to benchmark their own organisation's formal-benefit coverage and pay parity against the wider labour market.
- Researchers and journalists covering gender and employment in India could use the dashboard as a starting reference point for exploring specific states or demographic groups, without needing to repeat the underlying data cleaning and validation work themselves.
- State planning departments could use the correlation findings on Page 6 to weigh whether local job-creation policy or education policy is likely to have a larger effect on women's participation in their specific state.

14. Limitations

Every dataset and every dashboard has boundaries, and this project's are stated here directly rather than left implicit.

- The project is based on a single round of the PLFS survey (2023-24) and therefore describes the gap as it stands today rather than how it has changed over time; year-over-year trend analysis was outside this project's scope.
- The rural literacy dataset has no urban equivalent from the same source, so the literacy correlation reported in this project reflects rural literacy only, not a state's overall literacy rate.
- The per-capita GSDP dataset does not include Delhi, Chandigarh, Ladakh, or Puducherry, so state-level income comparisons involving these union territories are necessarily incomplete.
- The Reason_Not_Working and Efforts_To_Search_Work columns could not be decoded due to the absence of a verified official codebook and are excluded from all dashboard measures as a result, meaning the dashboard cannot currently show why non-working women say they are not working in their own stated terms, only through proxy measures such as domestic duty status.
- Some state-by-category breakdowns, particularly for smaller states and union territories, are based on relatively small sample sizes within the survey. These are shown on the dashboard but should be read with appropriate caution, and are flagged as such in the underlying methodology notes.
- The Avg Wage Gap and Formal Job % measures on the "Working, But How?" page could not be reproduced to the exact rupee or percentage point through an independent recalculation from the raw earnings and benefit columns alone, though the direction and approximate scale of both findings were confirmed. The dashboard's own DAX-calculated figures, built and tested within Power BI directly, remain the source of truth for these two measures.

15. Recommendations

The dashboard's closing page moves deliberately from analysis to action, offering four priorities that follow directly from the data rather than from general assumptions about the problem. They are reproduced and explained here in full.

15.1 Fix the marriage penalty

Marriage cuts women's workforce participation from roughly two-thirds to about one-third, a pattern confirmed by two independent methods in this project. The most direct response is not a single policy but a set of supports aimed specifically at the point of marriage and shortly after: accessible childcare, flexible and part-time formal job options, and structured re-entry programmes for women who have paused their careers. Because this is the single largest driver found anywhere in the data, it should be treated as the first priority rather than one initiative among many.

15.2 Skill young women early

Nearly one in two young women, against roughly one in ten young men, is currently NEET. Skilling and vocational training delivered before the typical age of marriage reaches women while they still have full access to the workforce, rather than after participation has already dropped. Timing matters here as much as the availability of the programme itself.

15.3 Recognise women's role at home

Eighty-three percent of household heads recorded in the survey are men, and 99 percent of recorded spouses are women. Women are not absent from the household economy — they are structurally invisible within how it is recorded. Any programme aimed at economic inclusion should account for unpaid domestic and care work explicitly, rather than treating a woman's absence from paid employment statistics as evidence of an absence of economic contribution.

15.4 Fix the worst states first

Female participation ranges from roughly 14 percent in Bihar to around 60 percent in Arunachal Pradesh. Given that state income correlates more strongly with participation than literacy does, funding that creates actual job access — rather than education spending alone — is likely to have the larger effect in the states currently lagging furthest behind, and should be prioritised there first.

These four priorities target the mechanisms driving the gap, not just its symptoms.

16. Conclusion

This project set out to answer a specific question — why fewer women work in India compared to men, and whether that gap changes with geography, education, and income — using real government survey data rather than assumption. The answer that emerged is more specific than the question itself. Geography and education both matter, but neither comes close to marriage as a driver: a woman's likelihood of working falls sharply and consistently the moment she marries, while a man's does not move at all. This finding held up under three separate analytical approaches, which is a stronger basis for confidence than any single chart could offer on its own.

Beyond that headline finding, the data paints a fuller picture worth holding onto: the gap is wider in Indian cities than in villages, nearly half of young women are neither working nor studying, and women who do contribute to their households are still overwhelmingly recorded in relation to a man rather than as an economic participant in their own right. None of these patterns are visible from national averages alone — they only appear once the data is broken down by state, age, marital status, and household role, which is precisely what this dashboard was built to do.

The project also demonstrates a complete, real-world analytics workflow: sourcing and cleaning a genuinely large government dataset in SQL, transforming and decoding it correctly in Power Query, building a validated set of DAX measures, and presenting the result as an interactive, story-driven dashboard rather than a static report. Every headline number carried through to this report was checked against the raw underlying data before being trusted, and the project's own honest documentation of what could not be verified is, in its own way, as important a result as the findings themselves.

17. Future Scope

A number of enhancements were identified during the build but placed outside this project's current scope, either due to time or because they depend on data not used here. They are recorded as a roadmap rather than left unmentioned.

- Custom top navigation bar and further refinement of the Story Mode bookmark sequence, once every dashboard page reaches its final, fully polished state.
- Rebuilding the Page 5 insight banner and Marriage Effect callout images using Segoe UI directly on a Windows machine, since the automated environment used to generate the current versions substitutes a similar but not identical font.
- A two-time-period comparison using an earlier PLFS round, such as 2019-20, to show whether the gender gap is widening, narrowing, or holding steady over time — something the current single-round dataset cannot answer on its own.
- A dedicated Power BI mobile phone layout, so the dashboard remains fully usable on a smaller screen rather than only on desktop.
- A one-page static PDF executive summary export, for audiences who want the top findings at a glance without opening the full interactive file.
- Custom hover tooltip pages showing additional mini-statistics on selected charts, to add depth for viewers who want to go further without cluttering the main page layout.
- Locating or reconstructing a verified codebook for the Reason_Not_Working and Efforts_To_Search_Work columns, which would allow the "Why Not?" page to show women's own stated reasons for not working directly, rather than through proxy measures.

18. References

- Periodic Labour Force Survey (PLFS), July 2023 – June 2024. Ministry of Statistics and Programme Implementation (MoSPI), National Statistics Office, Government of India. Accessed via the official microdata portal, microdata.gov.in.
- State/UT-wise Rural Literacy Rate, India, 2019-20 to 2023-24. Open Government Data (OGD) Platform India, data.gov.in.
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- International Monetary Fund (IMF), 2018. Statistic cited on the dashboard's opening page regarding the potential economic impact of closing India's gender employment gap.